Can an LLM Drive Real DRC Sign-off?

Connecting an LLM to deterministic EDA execution

Shivaram Mysore · Vyges · DAC 2026

Shivaram Mysore
Founder & CEO
shivaram@vyges.com
San Francisco Bay Area

The Problem

Reasoning is not execution

```
LLM
"Run DRC on this layout"
        ↓
        ?
        ↓
Real DRC Engine
```

We deliberately did not ask the model to become a DRC engine.

And yes—one capable model can probably be prompted to call one tool. That is not the problem we were trying to solve.

Architecture: The model reasons. The engine executes.

```
  user request
       |
       ▼
local LLM  —— selects the tool
       |
       ▼
structured MCP call
       |
       ▼
real DRC engine  —— computes
       |
       ▼
local LLM  —— reports what it found
```

Reason → Execute → Observe

```
                  ┌──────────────┐
                  |  LLM         |
                  |  + MCP       |
                  └──────┬───────┘
                         |
          ┌──────────────┼──────────────┐
          ▼              ▼              ▼
       DRC            STA            LVS
          |              |              |
  Self-description Self-description  Self-description
```

No bespoke orchestration per engine

Actual Run

No training. No prompt library. No hard-coded DRC flow.

```
> Check this layout and report whether it is DRC-clean.

  {"tool": "drc", "args": ["check", "spi_host_lite.gds",
                           "--rules", "sky130-core.drc",
                           "--top", "spi_host_lite"]}

  vyges-drc → CLEAN ✓  (no violations)

  "The layout is DRC-clean with 0 violations."
```

A locally hosted **SemiKong-8B**. A real macro from a **taped-out** chip. The model selected the DRC tool and generated the structured invocation. The engine returned the verdict. The model reported the engine's result, not its own.

The Takeaways

LLM reasoning. Deterministic EDA.

- **Model-agnostic** — change the model, not the execution layer
- **Tool-grounded** — the engine is the source of truth

LLM is not the sign-off engine. It is the driver.

- Live demo (GPT 4.x): https://vyges.github.io/vyges-loom-testbench/
- Paper — https://vyges.com/about/publications/
- Vyges CLI installation - https://docs.vyges.com/installation.html
- **shivaram@vyges.com** | vyges.com | YouTube